

LAB EXPERIMENT

ITA0506-COMPUTER VISION

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EXPNO:10

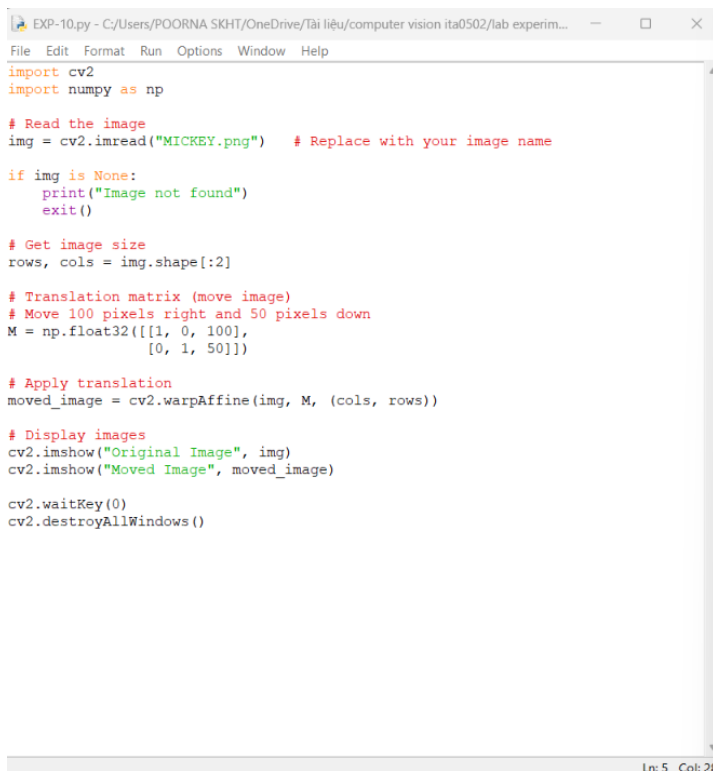
Aim:

To move an image from one position to another using image translation technique in Python and OpenCV.

Algorithm:

1. Import the required OpenCV library.
2. Read the input image.
3. Get the image dimensions.
4. Create a translation matrix with required x and y shift values.
5. Apply translation using cv2.warpAffine().
6. Display the original and moved images.
7. Close all display windows.

Code:



```
EXP-10.py - C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experim...
File Edit Format Run Options Window Help
import cv2
import numpy as np

# Read the image
img = cv2.imread("MICKEY.png") # Replace with your image name

if img is None:
    print("Image not found")
    exit()

# Get image size
rows, cols = img.shape[:2]

# Translation matrix (move image)
# Move 100 pixels right and 50 pixels down
M = np.float32([[1, 0, 100],
                [0, 1, 50]])

# Apply translation
moved_image = cv2.warpAffine(img, M, (cols, rows))

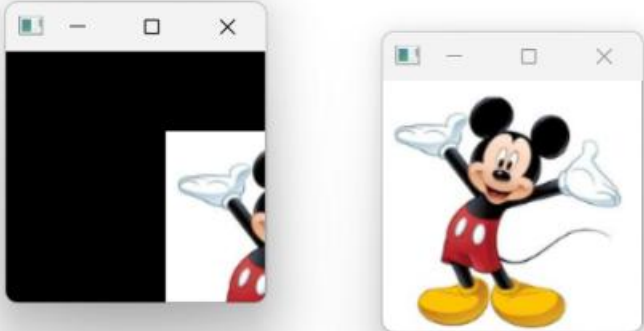
# Display images
cv2.imshow("Original Image", img)
cv2.imshow("Moved Image", moved_image)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

Ln: 5 Col: 28

OUTPUT :

```
File Edit Shell Debug Options Window Help
Python 3.13.3 (tags/v3.13.3:6280bb5, Apr 8 2025, 14:47:33) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-10.py
```



Ln: 5 Col: 0

RESULT: The image was successfully moved from one position to another using image translation in OpenCV.